

The Johansen Procedure: Full Derivation

Companion notes for Unit 7

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Table of contents

1	Overview	2
2	Setup and Notation	2
2.1	The VECM	2
2.2	Partialling out short-run dynamics (FWL)	2
2.3	Sample moments	3
3	The Gaussian Log-Likelihood	3
4	Profiling Out Σ_u	3
5	Profiling Out α Given β	4
5.1	Quadratic-in- α form	4
5.2	Completing the square	4
5.3	Why $\hat{\alpha} = CB^{-1}$ minimizes the determinant	4
5.4	The concentrated likelihood in β	5
6	Reducing to a Generalized Eigenvalue Problem	5
6.1	Two determinant identities	5
6.2	Normalization and the reduced problem	5
6.3	First-order condition	6
6.4	Diagonalization and the generalized eigenvalue problem	6
6.5	Picking the maximizer	6
7	Determinants in Eigenvalue Form	7
7.1	Unrestricted residual covariance	7
7.2	Rank- r restricted residual covariance	7
8	The LR Rank Tests	8
8.1	The likelihood ratio	8
8.2	Trace test	8
8.3	Maximum eigenvalue test	8
9	Summary of the Full Argument	9

1 Overview

These notes fill in every step of the derivation underlying the Johansen procedure. The slides give the storyline; here we give the algebra. The target audience is a student who is comfortable with multivariate calculus and matrix algebra but has never seen profile-likelihood arguments on matrix parameters.

The argument has four moves, and every Johansen result on the slides is a consequence of one of them:

1. **Concentrate out short-run dynamics** by partialling — the FWL step. This reduces the VECM to a static-looking long-run regression in residualized variables.
2. **Profile out** Σ_u . With Gaussian errors, for any fixed Π , the MLE of Σ_u is the sample covariance of the residuals. Substituting back, the log-likelihood becomes $-\frac{T}{2} \log |\widehat{\Sigma}_u(\Pi)| + \text{const.}$
3. **Profile out** α given β . With $\Pi = \alpha\beta'$, $\widehat{\Sigma}_u(\alpha, \beta)$ is quadratic in α , so we can complete the square. The minimizer is $\hat{\alpha} = S_{01}\beta(\beta'S_{11}\beta)^{-1}$.
4. **Reduce the β -problem to a generalized eigenvalue problem.** After plugging $\hat{\alpha}$ in and using two standard determinant identities, the constrained minimization in β becomes the eigenproblem

$$S_{10}S_{00}^{-1}S_{01}v = \lambda S_{11}v.$$

Once we are at the eigenvalue problem, the LR rank tests fall out mechanically: every $\widehat{\Sigma}$ is a product of $(1 - \hat{\lambda}_i)$ factors, so the LR statistic is a sum of $\log(1 - \hat{\lambda}_i)$.

Throughout, $x_t \in \mathbb{R}^m$ is $I(1)$, the cointegrating rank is r , and we work conditionally on the lagged differences and any deterministic terms.

2 Setup and Notation

2.1 The VECM

The VECM representation of a $\text{VAR}(p)$ in levels is

$$\Delta x_t = \Pi x_{t-1} + \sum_{i=1}^{p-1} \Gamma_i \Delta x_{t-i} + u_t, \quad u_t \sim N(0, \Sigma_u). \quad (1)$$

Cointegration imposes $\text{rank}(\Pi) = r < m$, factored as $\Pi = \alpha\beta'$ with $\alpha, \beta \in \mathbb{R}^{m \times r}$ both of full column rank.

2.2 Partialling out short-run dynamics (FWL)

Let $W_t = (\Delta x'_{t-1}, \dots, \Delta x'_{t-p+1})'$. Define the FWL residuals

$$\begin{aligned} R_{0t} &= \Delta x_t - \widehat{\mathbb{E}}[\Delta x_t | W_t], \\ R_{1t} &= x_{t-1} - \widehat{\mathbb{E}}[x_{t-1} | W_t], \end{aligned}$$

where $\widehat{\mathbb{E}}[\cdot | W_t]$ denotes the linear projection onto the span of W_t (in practice, OLS).

By the Frisch–Waugh–Lovell theorem applied to (Equation 1), the OLS estimator of Π from the original system is identical to the OLS estimator from the residualized regression

$$R_{0t} = \Pi R_{1t} + u_t, \quad (2)$$

and the residuals u_t are the same. Under Gaussianity, the conditional log-likelihood for Π given the lagged differences factors as the marginal log-likelihood of (Equation 2) plus a Π -free piece. So all subsequent likelihood maximization can be done on (Equation 2) alone — this is what justifies treating the problem as a static regression in R_{0t}, R_{1t} .

2.3 Sample moments

Define

$$S_{ij} = \frac{1}{T} \sum_{t=1}^T R_{it} R'_{jt}, \quad i, j \in \{0, 1\}.$$

These are $m \times m$ matrices; S_{00} and S_{11} are symmetric positive-definite (with probability one, for T large enough), and $S_{10} = S'_{01}$.

3 The Gaussian Log-Likelihood

Stack the residuals as columns, $E = [u_1, \dots, u_T]$. Then $\sum_t u_t u'_t = EE'$. Under $u_t \sim N(0, \Sigma_u)$ i.i.d., the log-likelihood from (Equation 2) is

$$\ell(\Pi, \Sigma_u) = -\frac{T}{2} \log |\Sigma_u| - \frac{1}{2} \text{tr}(\Sigma_u^{-1} EE') + \text{const},$$

where $E = E(\Pi)$ depends on Π through $u_t = R_{0t} - \Pi R_{1t}$.

4 Profiling Out Σ_u

Fix Π and maximize over Σ_u . The derivative of the log-likelihood with respect to Σ_u^{-1} yields

$$\widehat{\Sigma}_u(\Pi) = \frac{1}{T} E(\Pi) E(\Pi)' = \frac{1}{T} \sum_{t=1}^T (R_{0t} - \Pi R_{1t})(R_{0t} - \Pi R_{1t})'. \quad (3)$$

Plugging this back in, the trace term simplifies: with $\widehat{\Sigma}_u := \widehat{\Sigma}_u(\Pi)$,

$$\text{tr}(\widehat{\Sigma}_u^{-1} EE') = \text{tr}((T\widehat{\Sigma}_u)^{-1}(T\widehat{\Sigma}_u)) = \text{tr}(I_m) = m,$$

which is constant in Π . (The factor T comes from $EE' = T\widehat{\Sigma}_u$.) So the profiled log-likelihood reduces to

$$\ell(\Pi) = -\frac{T}{2} \log |\widehat{\Sigma}_u(\Pi)| + \text{const}.$$

Consequence. The constrained MLE problem is equivalent to

$$\min_{\Pi: \text{rank}(\Pi)=r} |\widehat{\Sigma}_u(\Pi)|.$$

This is a determinant minimization in a single matrix parameter Π of restricted rank.

5 Profiling Out α Given β

5.1 Quadratic-in- α form

Expand (Equation 3) using $\Pi = \alpha\beta'$:

$$\widehat{\Sigma}_u(\Pi) = S_{00} - \Pi S_{10} - S_{01}\Pi' + \Pi S_{11}\Pi'. \quad (4)$$

(Verify by writing $EE'/T = S_{00} - \Pi S_{10} - S_{01}\Pi' + \Pi S_{11}\Pi'$ from the definition of S_{ij} .)

Substitute $\Pi = \alpha\beta'$ and define

$$C := S_{01}\beta, \quad B := \beta' S_{11}\beta.$$

Then

$$\widehat{\Sigma}_u(\alpha, \beta) = S_{00} - \alpha C' - C\alpha' + \alpha B\alpha'. \quad (5)$$

Here B is symmetric positive-definite (since S_{11} is and β has full column rank), so $B^{1/2}$ is well-defined.

5.2 Completing the square

The cross-terms in (Equation 5) are linear in α and the $\alpha B\alpha'$ term is quadratic. Complete the square:

$$\widehat{\Sigma}_u(\alpha, \beta) = \underbrace{S_{00} - CB^{-1}C'}_{=: \Sigma_*(\beta)} + (\alpha - CB^{-1})B(\alpha - CB^{-1})'. \quad (6)$$

This is straightforward to verify by expanding the second term: $\alpha B\alpha' - \alpha BB^{-1}C' - CB^{-1}B\alpha' + CB^{-1}BB^{-1}C' = \alpha B\alpha' - \alpha C' - C\alpha' + CB^{-1}C'$, which combines with $S_{00} - CB^{-1}C'$ to recover (Equation 5).

5.3 Why $\hat{\alpha} = CB^{-1}$ minimizes the determinant

Let $L := (\alpha - CB^{-1})B^{1/2}$, an $m \times r$ matrix. Then (Equation 6) reads

$$\widehat{\Sigma}_u(\alpha, \beta) = \Sigma_*(\beta) + LL'.$$

We need: $|\Sigma_* + LL'| \geq |\Sigma_*|$, with equality iff $L = 0$. This is the matrix-determinant lemma applied to a rank- r update:

$$|\Sigma_* + LL'| = |\Sigma_*| \cdot |I_r + L'\Sigma_*^{-1}L|. \quad (7)$$

Since Σ_* is positive-definite (Σ_* is the residual covariance from regressing R_{0t} on $\beta' R_{1t}$, which is positive-definite under non-collinearity), $L'\Sigma_*^{-1}L$ is positive semi-definite, so $|I_r + L'\Sigma_*^{-1}L| \geq 1$ with equality iff $L = 0$, i.e. $\alpha = CB^{-1}$. Therefore

$$\hat{\alpha}(\beta) = CB^{-1} = S_{01}\beta(\beta' S_{11}\beta)^{-1}. \quad (8)$$

Remark. Equation (@eq-alphahat) has a transparent interpretation: $\hat{\alpha}(\beta)$ is the OLS coefficient from regressing R_{0t} on the r -dimensional regressor $\beta' R_{1t}$. This is exactly what FWL would tell you to do once β is fixed.

5.4 The concentrated likelihood in β

Plug $\hat{\alpha}(\beta)$ back into (Equation 6): the second term vanishes, leaving

$$\widehat{\Sigma}_u(\beta) = \Sigma_*(\beta) = S_{00} - S_{01}\beta(\beta'S_{11}\beta)^{-1}\beta'S_{10}. \quad (9)$$

The MLE problem now reads

$$\min_{\beta \in \mathbb{R}^{m \times r}, \beta \text{ full column rank}} |\widehat{\Sigma}_u(\beta)|.$$

6 Reducing to a Generalized Eigenvalue Problem

6.1 Two determinant identities

We use:

Lemma (Determinant rewrite). *For $A \in \mathbb{R}^{m \times m}$ invertible and $B \in \mathbb{R}^{m \times m}$,*

$$|A - B| = |A| |I - A^{-1}B|.$$

Lemma (Sylvester's determinant identity). *For $U \in \mathbb{R}^{m \times r}$ and $V \in \mathbb{R}^{r \times m}$,*

$$|I_m + UV| = |I_r + VU|.$$

Apply the first lemma to (Equation 9) with $A = S_{00}$, $B = S_{01}\beta(\beta'S_{11}\beta)^{-1}\beta'S_{10}$:

$$|\widehat{\Sigma}_u(\beta)| = |S_{00}| \cdot |I_m - S_{00}^{-1}S_{01}\beta(\beta'S_{11}\beta)^{-1}\beta'S_{10}|.$$

Apply Sylvester's identity with $U = -S_{00}^{-1}S_{01}\beta$ and $V = (\beta'S_{11}\beta)^{-1}\beta'S_{10}$:

$$|I_m - S_{00}^{-1}S_{01}\beta(\beta'S_{11}\beta)^{-1}\beta'S_{10}| = |I_r - (\beta'S_{11}\beta)^{-1}\beta'S_{10}S_{00}^{-1}S_{01}\beta|.$$

Combining,

$$|\widehat{\Sigma}_u(\beta)| = |S_{00}| \cdot |I_r - (\beta'S_{11}\beta)^{-1}\beta'S_{10}S_{00}^{-1}S_{01}\beta|. \quad (10)$$

6.2 Normalization and the reduced problem

The decomposition $\Pi = \alpha\beta'$ is identified only up to right multiplication of β by an invertible $r \times r$ matrix (since $\Pi = (\alpha M^{-1})(\beta M)' = \tilde{\alpha}\tilde{\beta}'$ for any $M \in GL_r$). Impose the normalization

$$\beta'S_{11}\beta = I_r. \quad (11)$$

Under this normalization, (Equation 10) collapses to

$$|\widehat{\Sigma}_u(\beta)| = |S_{00}| \cdot |I_r - \beta'A\beta|, \quad A := S_{10}S_{00}^{-1}S_{01}.$$

Note A is symmetric positive semi-definite. Minimizing $|\widehat{\Sigma}_u(\beta)|$ is equivalent (since $|S_{00}|$ is constant) to *maximizing* $|\beta'A\beta|$ subject to (Equation 11). This is the canonical-correlation extremal problem:

$$\boxed{\max_{\beta \in \mathbb{R}^{m \times r}} |\beta'A\beta| \quad \text{s.t.} \quad \beta'S_{11}\beta = I_r.} \quad (12)$$

6.3 First-order condition

Write the Lagrangian for $\log |\beta' A \beta|$ subject to the normalization, with multiplier matrix $\Lambda \in \mathbb{R}^{r \times r}$:

$$\mathcal{L}(\beta, \Lambda) = \log |\beta' A \beta| - \text{tr}(\Lambda (\beta' S_{11} \beta - I_r)).$$

Differentiating $\log |\beta' A \beta|$: by Jacobi's formula, $d \log |M| = \text{tr}(M^{-1} dM)$ for any invertible M . With $M(\beta) = \beta' A \beta$,

$$dM = (d\beta)' A \beta + \beta' A (d\beta).$$

By cyclicity of the trace and symmetry of A ,

$$\text{tr}(M^{-1} dM) = 2 \text{tr}(M^{-1} \beta' A (d\beta)) = 2 \text{tr}(A \beta M^{-1} (d\beta)')^\top.$$

Differentiating the constraint similarly, the FOC reads

$$A \beta (\beta' A \beta)^{-1} = S_{11} \beta \Lambda$$

(up to a factor of 2 absorbed into Λ). Right-multiplying by $\beta' A \beta$ gives

$$A \beta = S_{11} \beta \Gamma, \quad \Gamma := \Lambda (\beta' A \beta). \quad (13)$$

6.4 Diagonalization and the generalized eigenvalue problem

The FOC (Equation 13) can be rotated. Replace β by βP for any $P \in GL_r$: then $A(\beta P) = S_{11}(\beta P)(P^{-1} \Gamma P)$, so we may pick P to diagonalize Γ . Choose P so that $P^{-1} \Gamma P = \text{diag}(\lambda_1, \dots, \lambda_r)$, and write the columns of βP as $v_1, \dots, v_r$. Then (Equation 13) reads, column by column,

$$A v_i = \lambda_i S_{11} v_i, \quad i = 1, \dots, r,$$

i.e.,

$$\boxed{S_{10} S_{00}^{-1} S_{01} v = \lambda S_{11} v.} \quad (14)$$

This is the **generalized eigenvalue problem** at the heart of Johansen's method. Equivalently, multiplying by S_{11}^{-1} , the eigenvalues are those of $S_{11}^{-1} S_{10} S_{00}^{-1} S_{01}$.

Order the eigenvalues $1 \geq \hat{\lambda}_1 \geq \hat{\lambda}_2 \geq \dots \geq \hat{\lambda}_m \geq 0$ and let $v_1, \dots, v_m$ be the corresponding eigenvectors normalized so $v_i' S_{11} v_j = \delta_{ij}$.

6.5 Picking the maximizer

Under (Equation 11), $\beta' A \beta$ has eigenvalues equal to a subset of $\hat{\lambda}_1, \dots, \hat{\lambda}_m$ (whichever eigenvectors β spans). To *maximize* $|\beta' A \beta| = \prod_{i=1}^r \mu_i$ (with μ_i the eigenvalues of $\beta' A \beta$), pick the largest r eigenvalues:

$$\hat{\beta}_r = [v_1, v_2, \dots, v_r], \quad \hat{\beta}_r' A \hat{\beta}_r = \text{diag}(\hat{\lambda}_1, \dots, \hat{\lambda}_r).$$

The corresponding $\hat{\alpha}$ comes from (Equation 8): $\hat{\alpha}_r = S_{01} \hat{\beta}_r$ (the normalization makes the $(\beta' S_{11} \beta)^{-1}$ factor disappear).

Remark (Canonical correlations). Define $C := S_{00}^{-1/2} S_{01} S_{11}^{-1/2}$. The eigenvalues $\hat{\lambda}_i$ are exactly the eigenvalues of $C' C$, i.e., the squared canonical correlations between R_{0t} and R_{1t} . The Johansen procedure is canonical correlation analysis between residualized differences and residualized levels — a fact worth stating to students because it demystifies what the eigenvalues mean (they are correlations, not abstract roots).

7 Determinants in Eigenvalue Form

Two determinants enter the LR statistic: the unrestricted $|\widehat{\Sigma}_m|$ (full-rank Π) and the rank- r restricted $|\widehat{\Sigma}_r|$.

7.1 Unrestricted residual covariance

For full-rank Π , the unconstrained MLE is OLS: $\widehat{\Pi} = S_{01}S_{11}^{-1}$. Plugging into (Equation 4),

$$\widehat{\Sigma}_m = S_{00} - S_{01}S_{11}^{-1}S_{10} - S_{01}S_{11}^{-1}S_{10} + S_{01}S_{11}^{-1}S_{11}S_{11}^{-1}S_{10} = S_{00} - S_{01}S_{11}^{-1}S_{10}.$$

Compute $|\widehat{\Sigma}_m|$ in eigenvalue form. Pre- and post-multiply by $S_{00}^{-1/2}$:

$$S_{00}^{-1/2}\widehat{\Sigma}_m S_{00}^{-1/2} = I_m - S_{00}^{-1/2}S_{01}S_{11}^{-1}S_{10}S_{00}^{-1/2} = I_m - CC',$$

where $C = S_{00}^{-1/2}S_{01}S_{11}^{-1/2}$ as in the remark above. The eigenvalues of CC' equal those of $C'C = S_{11}^{-1/2}S_{10}S_{00}^{-1}S_{01}S_{11}^{-1/2}$, and this matrix is similar to $S_{11}^{-1}S_{10}S_{00}^{-1}S_{01}$, whose eigenvalues are the Johansen eigenvalues $\hat{\lambda}_1, \dots, \hat{\lambda}_m$. Therefore

$$|I_m - CC'| = \prod_{i=1}^m (1 - \hat{\lambda}_i),$$

and taking determinants of both sides of the similarity transform,

$$|\widehat{\Sigma}_m| = |S_{00}| \prod_{i=1}^m (1 - \hat{\lambda}_i). \quad (15)$$

7.2 Rank- r restricted residual covariance

From (Equation 9) and the normalization (Equation 11),

$$\widehat{\Sigma}_r = S_{00} - S_{01}\hat{\beta}_r\hat{\beta}_r'S_{10}.$$

Pre- and post-multiply by $S_{00}^{-1/2}$:

$$S_{00}^{-1/2}\widehat{\Sigma}_r S_{00}^{-1/2} = I_m - CP_rC', \quad P_r := S_{11}^{1/2}\hat{\beta}_r\hat{\beta}_r'S_{11}^{1/2},$$

where the factorization absorbs the $S_{11}^{1/2}$ pieces into a clean projector. Specifically, set $W_r := S_{11}^{1/2}\hat{\beta}_r$, an $m \times r$ matrix. The normalization $\hat{\beta}_r'S_{11}\hat{\beta}_r = I_r$ gives $W_r'W_r = I_r$, so

$$P_r = W_rW_r'$$

is the *orthogonal projector* onto $\text{span}(W_r) \subset \mathbb{R}^m$.

P_r projects onto the top- r eigenspace of $C'C$. From (Equation 14), the eigenvectors v_i of the generalized problem satisfy $C'Cw_i = \hat{\lambda}_i w_i$ where $w_i = S_{11}^{1/2}v_i$ (verify: $S_{11}^{-1/2}S_{10}S_{00}^{-1}S_{01}S_{11}^{-1/2}(S_{11}^{1/2}v_i) = S_{11}^{-1/2}Av_i = S_{11}^{-1/2} \cdot \hat{\lambda}_i S_{11}v_i = \hat{\lambda}_i(S_{11}^{1/2}v_i)$). The columns of W_r are exactly $w_1, \dots, w_r$, the top- r eigenvectors of $C'C$.

Eigenvalues of CP_rC' . For each w_i with $\hat{\lambda}_i > 0$, the vector Cw_i is an eigenvector of CC' with the same eigenvalue $\hat{\lambda}_i$. Now apply CP_rC' to Cw_i :

- If $i \leq r$: $w_i \in \text{span}(W_r)$, so $P_r w_i = w_i$, hence $CP_r C'(Cw_i) = CP_r(C' C)w_i = \hat{\lambda}_i Cw_i$.
- If $i > r$: $w_i \perp \text{span}(W_r)$ (eigenvectors of distinct eigenvalues are orthogonal under the symmetric problem $C' C w = \lambda w$), so $P_r w_i = 0$, hence $CP_r C'(Cw_i) = 0$.

The non-zero eigenvalues of $CP_r C'$ are therefore exactly $\hat{\lambda}_1, \dots, \hat{\lambda}_r$, and the remaining $m - r$ eigenvalues are zero. Hence the eigenvalues of $I_m - CP_r C'$ are $1 - \hat{\lambda}_1, \dots, 1 - \hat{\lambda}_r, 1, \dots, 1$, and

$$|I_m - CP_r C'| = \prod_{i=1}^r (1 - \hat{\lambda}_i).$$

Putting it together,

$$\boxed{|\widehat{\Sigma}_r| = |S_{00}| \prod_{i=1}^r (1 - \hat{\lambda}_i).} \quad (16)$$

8 The LR Rank Tests

8.1 The likelihood ratio

The LR statistic for $H_0 : \text{rank}(\Pi) \leq r$ against the unrestricted alternative is

$$\text{LR}(r) = -2(\ell_r - \ell_m) = -T \log \frac{|\widehat{\Sigma}_m|}{|\widehat{\Sigma}_r|}.$$

Combining (Equation 15) and (Equation 16), the $|S_{00}|$ factors cancel:

$$\frac{|\widehat{\Sigma}_m|}{|\widehat{\Sigma}_r|} = \prod_{i=r+1}^m (1 - \hat{\lambda}_i),$$

so

$$\text{LR}(r) = -T \sum_{i=r+1}^m \log(1 - \hat{\lambda}_i).$$

8.2 Trace test

Compares rank r to the unrestricted ($r = m$) model:

$$\text{LR}_{\text{trace}}(r) = -T \sum_{i=r+1}^m \log(1 - \hat{\lambda}_i).$$

Tests $H_0 : \text{rank}(\Pi) \leq r$ against $H_1 : \text{rank}(\Pi) > r$. Under H_0 , all of $\hat{\lambda}_{r+1}, \dots, \hat{\lambda}_m$ should be near zero in population.

8.3 Maximum eigenvalue test

Compares rank r to rank $r + 1$:

$$\text{LR}_{\text{max}}(r, r + 1) = -T \log(1 - \hat{\lambda}_{r+1}).$$

Tests $H_0 : \text{rank}(\Pi) = r$ against $H_1 : \text{rank}(\Pi) = r + 1$.

Both statistics have non-standard limiting distributions (functionals of vector Brownian motion); critical values are tabulated in Johansen (1995) and depend on the deterministic specification (constant, trend, restricted constant, etc.).

9 Summary of the Full Argument

The four-move structure can be written as a single chain of equivalences:

$$\begin{aligned}
\max_{\Pi: \text{rank } \Pi = r} \ell(\Pi, \Sigma_u) &\stackrel{\text{profile } \Sigma_u}{\iff} \min_{\text{rank } \Pi = r} |\widehat{\Sigma}_u(\Pi)| \\
&\stackrel{\Pi = \alpha\beta'}{\iff} \min_{\alpha, \beta} |\widehat{\Sigma}_u(\alpha, \beta)| \\
&\stackrel{\text{profile } \alpha}{\iff} \min_{\beta} |\widehat{\Sigma}_u(\beta)| \\
&\stackrel{\text{determinant identities}}{\iff} \max_{\beta: \beta' S_{11} \beta = I_r} |\beta' A \beta| \\
&\stackrel{\text{Lagrangian} + \text{Jacobi}}{\iff} \text{top-}r \text{ generalized eigenvectors of } S_{11}^{-1} S_{10} S_{00}^{-1} S_{01}.
\end{aligned}$$

The eigenvalues $\hat{\lambda}_i$ produced by the eigenvalue problem parameterize every quantity of interest: $\hat{\beta}_r$ is the top- r eigenvectors, $\widehat{\Sigma}_r$ is $|S_{00}|$ times the product of $(1 - \hat{\lambda}_i)$ for $i \leq r$, and the LR rank statistics are sums of $\log(1 - \hat{\lambda}_i)$.

10 References (selected)

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